

Static Gesture Classification — Run Report

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Data paths

Training data path 1	/home/jestin/ThesisRepo/ML/NewTrainingData/Static · 365 trials
Total training paths	1 · 365 trials total
External test root	/home/jestin/ThesisRepo/ML/NewTestData/8_Jestin/Static
Report output dir	/home/jestin/ThesisRepo/ML/saved_reports

Sensor selection

Hands	left, right
Segments	palm_mid, thumb, index, middle, ring, pinky, wrist
Modalities	YPR, Accel, Flex
Sensor columns	176 channels
Classes	6: Double_Closed_Fist, Double_Okay, Double_Open_Palm, Double_Phone, Double_Pistol, Double_Spiderman

Preprocessing, features & split

Resample steps	90
Butterworth	off · cutoff 10.0 Hz · order 4 · fs 30.0 Hz
Normalisation	None
Feature mode	stats+fft → feature dim 9504
Test size · seed · CV folds	0.2 · random_state=42 · 10-fold stratified
Sample counts	train (post-aug)=292 · test=73

Augmentation (training only)

Enabled	no
Copies per train trial	1
Jitter	off · sigma=0.01
Amplitude scale	on · range=(0.8, 1.2)
Time warp	off · range=(0.9, 1.1)

Classifiers — cross-validation & hold-out test

Algorithm	CV mean (10-fold)	CV std	Hold-out test acc
SVM (Linear)	1.0000	0.0000	1.0000
Random Forest	1.0000	0.0000	1.0000
Logistic Regression	0.9897	0.0221	1.0000
LDA	0.9966	0.0103	1.0000
Voting Soft (SVM+RF+KNN+LogReg)	0.9966	0.0103	1.0000
Stacking (SVM+RF+KNN+LogReg)	0.9966	0.0103	1.0000
KNN	0.9490	0.0376	0.9863
SVM (RBF)	0.8905	0.0396	0.8904

Per-class hold-out F1

Class	SVM (Linear)	Random Forest	Logistic Regression	LDA	Voting (soft)	Stacking	KNN	SVM (RBF)	Support
Double_Closed_Fist	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.963	13
Double_Okay	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.786	12
Double_Open_Palm	1.000	1.000	1.000	1.000	1.000	1.000	1.000	0.667	12
Double_Phone	1.000	1.000	1.000	1.000	1.000	1.000	0.960	0.909	12
Double_Pistol	1.000	1.000	1.000	1.000	1.000	1.000	0.957	1.000	12
Double_Spiderman	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	12
macro avg	1.000	1.000	1.000	1.000	1.000	1.000	0.986	0.887	73

External test (NewTestData)

Algorithm	External test acc
Random Forest	1.0000
LDA	0.8387
SVM (RBF)	0.3226
SVM (Linear)	0.3226
KNN	0.3226
Logistic Regression	0.3226
Voting Soft (SVM+RF+KNN+LogReg)	0.3226
Stacking (SVM+RF+KNN+LogReg)	0.3226

Per-class external accuracy

[illegible]